Exam 2 — 92

Crystal Bowers
5:15 - 6:30pm

1) Beneficial safety features of SFRs include the large margin between melting and boiling temperature points for sodium (Na). The negative reactivity feedback allows for the coolant temperature to stop from continually increasing, the reactor can shutdown, and no operator action would have been needed. SFRs can operate at atmospheric pressures unlike LWRs and have good performance characteristics during transients.

6/6

2) We use alloy uranium for fuels because early on pure U fuel experiments showed excessive swelling (>30%) and failed due to FCMI until lowering the smear density to 75% was discovered. At temperatures on the lower end, (400-600°C) cavitation swelling dominates while the metal fuel is in an alpha state which included anisotropic swelling. Alloying metallic fuels can have benefits of stabilizing U with Zr to have more BCC gamma phases where swelling is more isotropic.

6/6

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3) Metallic fuel systems include a relatively low smear density (75%) and large plenum because early on there were a lot of metallic fuel failures due to FCMI (fuel cladding mechanical interaction) because of excessive fuel swelling (>30%). This put an outward force on cladding leading to cladding deformation or cladding rupture. The lower smear density and large plenum provides places for fission gas to go and decrease stress on the cladding. Overall these changes to the fuel design mitigated FCMI.

9/10

- bubble interconnection + gas release

why occurs + impacts Crystal B.

4) Constituent redistribution in metallic fuels means constituents/elements in the fuel moving from their original locations. In the case of metallic fuel, Zr redistributes because of the temperature gradient and chemical potential. You then get regions of higher concentrations of Zr near the fuel center, depleted Zr, and an as-fabricated region. This occurs due to diffusion due to the Soret effect and chemical potential. Zr moves up the thermal gradient / moves down the thermal gradient. Zirconium can help stabilize for phases to be gamma with the BCC structure.

13/14

- Wanted more impacts

Constituent redistribution can occur due to FCCI between the fuel and cladding. This means elements like Fe from the cladding go into the fuel and elements like Ce go from the fuel into the cladding. Phases with either the fuel or cladding's constituents can occur on either side. Impacts include areas with lower melting points which could be limiting during operation. FCCI occurs due to the chemical reactions as fission products are generated and interact with cladding along with the temperature gradient.

FCCI goes to

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5) Thermal conductivity is initially high at start of life for metal fuel. Thermal conductivity varies as a function of burnup in metallic fuels by early burnup/ earlier in life as an increase in Porosity degrades thermal conductivity. The gamma phase at the center of the fuel will have larger bubbles that can interconnect. Later in life, once those bubbles have interconnected and the maximum swelling has occurred and fission gas release has occurred, the sodium bond will then infiltrate the fuel leading to an increase in thermal conductivity; this is at a point when burnups are higher.

8/8

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6) For metallic fuel, FCCI is fuel cladding chemical interaction; FCCI is when chemical reactions are occurring between the fuel and cladding materials.

14/14

As fission occurs, fission products are generated that will interact with the cladding and have chemical reactions.

Primary fuel species participating in FCCI include Ce and lanthanide fission products. Primary Nd

Cladding species participating in FCCI include Fe + Cr. Phases on either side (fuel or cladding) can form with fuel elements (U, Pu, Zr) or cladding (Fe, Ni).

Adverse effects include local zones and effects where there is cladding wastage or the FCCI is impacting the depth of the cladding. Phases with lower melting points could form which could limit operational parameters.

In addition, mechanical properties of cladding could be impacted.

7)

Restructuring in mox occurs by there being a central void region, columnar grain growth region, equiaxed grain growth region, and as-sintered region. It occurs due to the steep temperature gradient in mox since it has low thermal conductivity, small pores are very mobile and will move up the temperature gradient. Lenticular / Intermediate pores will travel the temperature gradient leaving streaks. Voids and vacancy migration due to the temperature gradient leads towards the center of the fuel leads to the central void forms. the columnar grain growth region is formed due to an evaporation-condensation mechanism as pores have an even steeper temperature gradient and higher pressures will be on the hotter side of the pore.

12/12

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>) cont. Vapor Pressure is high on the hot side of the lenticular pore and matter evaporates from the hot face and condensates on the cold face leading to inverse displacement. The as-sintered region has lower temperatures, which results in less impacts to its structure on the periphery of the fuel.

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8) Pu concentration in a MOX pin becomes enriched around the central void.

7/10 O concentration is higher near the rim / periphery of the MOX pin. Near the central void, the increased concentration of Pu leads to locally higher fission rates.

- why? Pu agglomerates?

Wanted a little more

9) JOG is Joint Oxide Grain and it's an oxide layer that forms on the surface of the MOX fuel and between the fuel and cladding. It occurs because

11/12 volatiles move to the cooler temperatures at the periphery of the fuel. There's excess O due to the preferential formation of nobles. Impacts include acting as a mechanical buffer decreasing FCMI impacts, increasing thermal transport since an oxide would have higher thermal conductivity than the He gap, and contributing a lot to FCMI.

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10) Sodium corrosion occurs in SFRs via (2) types. The first type is dissolution of alloying elements due to the exposure to sodium (Na). The other type is due to chemical reactions with impurities due to exposure to Na.

6/8

Limiting the concentration of oxygen can help prevent corrosion impacts as oxygen can increase the potential solubility of some constituents / elements.

- thermal gradient mass transfer?
- impurities ↑ solubility